Use Case - PCA

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Background & Literature

Principal component analysis (PCA) is a linear dimensionality reduction technique commonly used in exploratory data analysis or data preprocessing. It transforms high-dimensional, correlated variables into a new coordinate system of fewer, uncorrelated variables, also known as principal components (PCs). PCA works by centering (and rescaling) data, calculating the covariance or correlation matrix, and findings its eigenvalues and eigenvectors. Eigenvectors represent the directions of the PCs, and eigenvalues quantify the amount of variance explained by each corresponding PC.

Uncertainty quantification (UQ) in PCA is multifaceted. One may ask how reliable the eigenvalues are, loadings (variable contributions to each component), scores (projection of each sample onto that component), or the number of retained components, given noisy or limited data. Most existing approaches treat PCA results as statistical estimators and quantify uncertainty through their sampling distributions. Uncertainties are typically described using a confidence interval: a narrow interval is interpreted as high certainty, whereas a wide interval implies low certainty.

Main PCA Quantities and UQ Approaches

Quantity	Main UQ Approach	Citations
Loadings	Bootstrap, broken-stick CIs, PPCA	Timmerman et al. (2007); Peres-Neto et al. (2003); Rahoma et al. (2021); Tipping et al. (1999)
Eigenvalues	Bootstrap CIs, PPCA	Dorabiala et al. (2023); Kalantan et al. (2025); Tipping et al. (1999)
Scores	Bootstrap scores	Castura et al. (2021)
Number of components / factors	Subsampling CIs, parallel analysis	Jha & Barnett (2022); Franklin et al. (1995)

In applied settings, choosing the number of PCs is one of the most important objectives in PCA (and also factor analysis). Underspecifying number of PCs would lead to loss of information, and overspecifying number of PCs can weaken signal strengths and reduce interpretability. The number of retained PCs is not just a random output; it is determined by the eigen-structure of the covariance matrix. Most existing UQ approaches are local and marginal, focusing on individual quantities such as eigenvalues or loadings, which may miss important structural ambiguity. For example, two datasets may have identical eigenvalues but have completely different loading structures. Therefore, instead of focusing on individual parameters, it is important to evaluate uncertainty of the underlying space of the eigenstructure.

Research Question

How can we quantify uncertainty in PCA outputs by accounting for variability in the entire eigensystem rather than individual quantities?

Data

Using the spectral decomposition theorem, we construct population covariance matrices by pre-specifying their eigenvalues. Two covariance scenarios are considered: a fragile condition and a stable condition. In the fragile condition, Σ_{Fragile} is constructed with small eigengaps among the leading eigenvalues, for example $\lambda_1 = 6.0, \lambda_2 = 5.9, \ldots$, so that the leading eigenspaces are expected to be more sensitive to sampling perturbation. In contrast, Σ_{Stable} is constructed with well-separated eigenvalues, for example $\lambda_1 = 6.0, \lambda_2 = 2.0, \ldots$, yielding more identifiable principal directions.

For each scenario, an orthogonal matrix Q is generated by drawing a $p \times p$ matrix with independent standard normal entries and applying QR decomposition. The covariance matrix is then formed as

$$\Sigma = Q \Lambda Q^\top,$$

where Λ is the diagonal matrix of pre-specified eigenvalues. Multivariate normal data are sampled from each population:

$$X \sim N(0, \Sigma_{\text{Fragile}}), \qquad Y \sim N(0, \Sigma_{\text{Stable}}),$$

with dimension $p = 20$. Empirical correlation matrices are estimated from the simulated data and used as the starting point for the proposed uncertainty quantification procedure. To evaluate the impact of sampling variability, simulations are repeated across sample $n \in 50, 100, 200, 500, 1000, 2000, 5000$.

This design allows us to evaluate whether the proposed procedure detects greater subspace instability when eigengaps are narrow and whether this instability decreases as sample size increases.

Methods

The starting point of the uncertainty quantification procedure is the observed empirical correlation matrix, $\hat{R}$. The population correlation matrix R will be treated as unknown, all inference is conducted relative to the uncertainty around $\hat{R}$. A 95% MCMC perturbation cloud of plausible correlation matrices, $\{\hat{R}^{(b)}\}_{b=1}^{B}$, will be generated around $\hat{R}$, with $B = 20{,}000$. PCA will then be performed on each $\hat{R}^{(b)}$, and the resulting eigenvalues, eigenvectors, and loadings will be summarized to characterize uncertainty in the PCA structure.

The primary objective is to determine the number of principal components to retain while accounting for uncertainty. Rather than selecting components based only on eigenvalues from a single PCA, we frame component selection as identifying a low-dimensional subspace that remains stable under perturbations of the observed correlation structure. For each candidate dimension k, the leading k-dimensional eigenspace from perturbation b is represented by

$$P_k^{(b)} = V_k^{(b)} V_k^{(b)\top},$$

where $V_k^{(b)}$ contains the first k eigenvectors of $\hat{R}^{(b)}$.

Stability is assessed by measuring the deviation of perturbation-based subspaces from the reference subspace obtained from $\hat{R}$. Specifically, for each k, we compute

$$d^{(b,k)} = \|P_k^{(b)} - P_k^{(0)}\|_F,$$

where $P_k^{(0)}$ is the projection matrix derived from the observed $\hat{R}$. The distribution of $d^{(b,k)}$ across perturbations will be summarized using boxplots. A subspace is considered stable if this distribution remains small and concentrated. The number of principal components retained is selected as the largest k for which the corresponding subspace remains stable.

As a comparison, a conventional PCA workflow will be applied to the single observed correlation matrix $\hat{R}$. The number of retained components will be determined using standard tools such as the scree plot and eigenvalue-based criteria, and results will be compared with the uncertainty-aware selection based on the perturbation cloud.

Table Shells

Table 1: Subspace Stability Summary by Dimension k (Note: k=5 refers to the subspace spanned by the first 5 PCs). This helps us to choose number of PCs to retain.

k	Median distance	Max distance	(2.5th, 97.5th percentile) distance
1			
2			
3			
4			
5			

References

1. Timmerman, M., Kiers, H., & Smilde, A. (2007). Estimating confidence intervals for principal component loadings: a comparison between the bootstrap and asymptotic results.. *The British journal of mathematical and statistical psychology, 60 Pt 2*, 295-314. https://doi.org/10.1348/000711006x109636
2. Peres-Neto, P., Jackson, D., & Somers, K. (2003). Giving Meaningful Interpretation to Ordination Axes: Assessing Loading Significance in Principal Component Analysis. *Ecology, 84*, 2347-2363. https://doi.org/10.1890/00-0634
3. Rahoma, A., Imtiaz, S., & Ahmed, S. (2021). Sparse principal component analysis using bootstrap method. *Chemical Engineering Science*. https://doi.org/10.1016/j.ces.2021.116890
4. Dorabiala, O., Aravkin, A., & Member, I. (2023). Ensemble Principal Component Analysis. *IEEE Access, 12*, 6663-6671. https://doi.org/10.1109/access.2024.3350984
5. Kalantan, Z., Alharbi, L., Al-Zahrani, M., & Binhimd, S. (2025). Robust Dimensionality Reduction: A Bootstrap-Based Evaluation of PCA with Applications in Nutritional and Environmental Sciences. *Contemporary Mathematics*. https://doi.org/10.37256/cm.6120256016
6. Castura, J., Rutledge, D., Ross, C., & Næs, T. (2021). Discriminability and uncertainty in principal component analysis (PCA) of temporal check-all-that-apply (TCATA) data. *Food Quality and Preference*, 104370. https://doi.org/10.1016/j.foodqual.2021.104370
7. Jha, C., & Barnett, I. (2022). Confidence Intervals for the Number of Components in Factor Analysis and Principal Components Analysis via Subsampling. **.
8. Franklin, S., Gibson, D., Robertson, P., Pohlmann, J., & Fralish, J. (1995). Parallel Analysis: a method for determining significant principal components. *Journal of Vegetation Science, 6*, 99-106. https://doi.org/10.2307/3236261
9. Michael E. Tipping, Christopher M. Bishop, Probabilistic Principal Component Analysis, *Journal of the Royal Statistical Society Series B: Statistical Methodology*, Volume 61, Issue 3, September 1999, Pages 611–622, https://doi.org/10.1111/1467-9868.00196